# Ahmed Hesham Salah

Data Science Student | AI, Machine Learning & Backend Development
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## PROFESSIONAL SUMMARY

Data Science student at Helwan National University with hands-on experience building end-to-end AI systems, including a Retrieval-Augmented Generation (RAG) assistant with a local LLM, CNN image classifiers, FastAPI backends, and interactive data dashboards. Completed ITI training in AI and Flutter. Available for internships and entry-level roles in AI, machine learning, and data science.

## TECHNICAL SKILLS

**Programming Languages:** Python, R, C++, Java, SQL, Dart, Solidity

**Generative AI & RAG:** Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Ollama, Prompt Engineering, Text Embeddings, Sentence-Transformers, Vector Databases (ChromaDB)

**Machine Learning & Deep Learning:** Machine Learning, Deep Learning, CNN Architectures, Transfer Learning, Image Classification, Computer Vision, Natural Language Processing, Model Explainability

**Data & Analytics:** Data Analysis, Data Visualization, Statistical Problem-Solving, Pandas, NumPy, Plotly Dash, Jupyter Notebook

**Backend Development:** FastAPI, RESTful API Design, Pydantic, SQLite, Redis Caching, Role-Based Access Control, Streamlit

**Mobile & Blockchain:** Flutter (Dart), Clean Architecture, Solidity Smart Contracts, Web3.py, Ethereum

**Tools & DevOps:** Git, GitHub, Docker, Docker Compose, Pytest (Automated Testing), Python Scripting

**Languages:** Arabic (Native), English (Professional Proficiency)

## PROJECTS

**RAG-Powered Document Assistant** 2026

**GitHub:** github.com/Ahmad-Salah-22/RAG-Powered-Document-Assistant-

*Full-stack AI application that lets users ask questions over a PDF document collection and receive grounded answers with document and page citations, running entirely on a local LLM.*

- Built a data ingestion pipeline with PyPDF page-level extraction, text cleaning, and overlapping chunking (800 characters, 150 overlap); generated 384-dimension embeddings with Sentence-Transformers (all-MiniLM-L6-v2) and stored them in a persistent ChromaDB vector database for cosine-similarity top-K retrieval.
- Developed a FastAPI backend with POST /query and GET /health endpoints, Pydantic request validation, and structured logging; loaded the vector store and embedding model once at startup using the lifespan handler to avoid per-request overhead.
- Integrated a local Ollama LLM (Llama 3 8B) with a strict grounding prompt so answers cite their source chunks and fall back to a safe response instead of hallucinating on out-of-scope questions.
- Created a Streamlit chat interface with source citations, loading states, and error handling; wrote Pytest tests (valid query, HTTP 422 validation, health check) and containerized the backend and frontend with Docker Compose.
- Evaluated the pipeline on 10 test questions: in-domain queries retrieved the correct source document and page, and out-of-domain queries returned a fallback message without fabricating answers.

**Tech Stack:** Python, FastAPI, ChromaDB, Sentence-Transformers, Ollama, Streamlit, Docker, Pytest

**Spotify Analytics Dashboard** Feb 2026 – May 2026

**GitHub:** github.com/seifalmaz/spotify-analytics-dashboard

- Built an interactive Plotly Dash dashboard analyzing 54,000+ Spotify tracks across genre analysis, audio feature relationships, popularity trends, explicit vs. clean content, and time-series insights, with KPI views and a responsive dark-themed UI.
- Designed a metadata recovery pipeline using the Spotify API and track_id values to retrieve missing release information, closing dataset gaps required for temporal analysis.
- Enriched the data by joining a secondary Spotify dataset with left joins and validation checks, recovering missing dates while preserving consistency; enabled year filters, decade trends, and genre evolution analysis.

**Tech Stack:** Python, Plotly Dash, Pandas, NumPy, Jupyter Notebook

**Online Course Certificate Registry (Blockchain)** Jan 2026 – Apr 2026
**GitHub:** github.com/Ahmad-Salah-22/CertificateProject

- Built a blockchain-based digital certificate registry where course completions are permanently recorded and publicly verifiable on-chain.
- Developed Solidity smart contracts with role-based access control (admin/user), batch operations, and emergency pause/resume functionality.
- Implemented a Python terminal application with admin and user menus, live blockchain event alerts, and transaction history scanning via Web3.py.
- Wrote automated security tests, a balance snapshot CSV exporter, and an admin dashboard script for system-wide analytics.

**Tech Stack:** Solidity, Python, Web3.py, Ethereum

**Waste Classifier – Intelligent Recycling System**
**GitHub:** github.com/seifalmaz/waste-classifier

- Built a deep learning image classification system that sorts waste into 5 categories to support smart recycling and environmental sustainability.

**Hand Gesture Recognition System**
**GitHub:** github.com/seifalmaz/gesture-recognition

- Designed and trained CNN-based classifiers using transfer learning to recognize gestures (e.g., peace, stop, help) from images and real-time video streams.
- Compared multiple CNN architectures to identify the best-performing model; built a GUI to visualize predictions, confidence scores, and model explainability.

**Task Manager API**
**GitHub:** github.com/ahmedelhossary22/Task-Management-System-Python-and-R-language-project

- Built a backend system for managing projects and tracking tasks with workflow validation and role-based access control.
- Integrated Redis caching and structured logging to improve performance and maintainability.

**Tech Stack:** Python, FastAPI, SQLite, Redis

## EDUCATION

**Helwan National University** Sep 2023 – Present
Bachelor's Degree, Computer Science and Technology
**Relevant Coursework:** Data Structures, Algorithm Design, Data Analysis, Big Data, Computer Networks, Operating Systems

## WORK EXPERIENCE

**Frontend Developer (Freelance)** 2024 – 2026

- Delivered responsive, client-facing web interfaces by translating design mockups into clean, functional front-end code.
- Collaborated directly with clients to refine features and met project deadlines.

**Team Lead, Talent Acquisition – Hires Lab** Jul 2024 – Sep 2025

- Led talent sourcing efforts focused on building diverse candidate pipelines and strengthening employer branding.
- Used AI-driven sourcing tools and data analytics to optimize the hiring process.

## CERTIFICATIONS & TRAINING

- AI 2 Training – Information Technology Institute (ITI), Summer 2026
- Flutter Training – Information Technology Institute (ITI), Summer 2025
- Flutter Diploma – Route Academy
- Deep Learning in Python – NVIDIA
- Natural Language Processing in Python – DataCamp
- Image Processing in Python – DataCamp
- Computer Vision and Intro to Deep Learning – Kaggle
- Python Essentials 1 – Cisco Networking Academy and Python Institute